Model Safety, Moderation, and Robustness – Comparison Sheet

Topic	Key Idea	Examples / Methods	Challenges
Adversarial Inputs	Inputs crafted to trick models	Modified text/images, confusing queries	Hard to detect subtle attacks; safety-critical risks (healthcare, finance)
Prompt Injection	Embedding malicious instructions in LLM prompts	Direct injection ('ignore rules…'), Indirect injection (hidden in docs/links)	Can override system rules; leaks confidential info
Jailbreaking LLMs	Bypassing safeguards to force unsafe outputs	Role-playing, obfuscation, hypotheticals	Leads to harmful or restricted content; difficult to fully block
Ethical Concerns	Responsible use of LLMs	Harmful info, bias, misinformation, cyber misuse	Accountability issues: developer vs. deployer vs. user
Content Filtering & Moderation	Preventing harmful outputs	- Rule-based (keywords) - Classifier-based - Human-in-loop - Context-aware filtering	Balancing false positives/negatives; cultural/legal variations
Training Techniques	Improve robustness & alignment	Adversarial training, bias reduction, data sanitization	Requires diverse datasets; may reduce creativity
Post-training Safeguards	Aligning model outputs	RLHF, safety fine-tuning	Costly, requires expert human feedback
Robustness Measures	Protect against misuse	Input validation, rate-limiting, monitoring	Attackers adapt quickly; system overhead
Evaluation & Testing	Ongoing safety checks	Red-teaming, stress tests, audits	New attack methods constantly emerging
Transparency & Trust	Make systems understandable	Explainable outputs, clear policies	Hard to balance openness with security